

Classification & Genesis of Metamorphic Rocks: Blueschist, Pseudotachylite, Hornfels & Skarn

Authoritative References: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapters 21, 22, 23 & 30)

Comprehensive Chapter Masterclass: Metamorphic Settings & Rock Nomenclature

Subduction High-P Metamorphism, Dynamic Fault Rocks & Frictional Melts, Thermal Aureoles & Metasomatic Skarns

1. The Spectrum of Metamorphic Environments

Metamorphic rocks are classified on the basis of their **texture, mineralogy, protolith chemistry, and geological setting of origin** (Winter, 2010, Chapters 21 & 22; IUGS Subcommittee on the Systematics of Metamorphic Rocks - SCMR). Four critical classes represent extreme end-members of physical regimes:

- **Blueschist (Subduction Zone High-P/Low-T):** Metabasic rocks metamorphosed under high lithostatic pressure and low temperature ($P > 6 \text{--} 10 \text{ kbar}$, $T = 200^{\circ}\text{--}450^{\circ}\text{C}$) in subducting oceanic slabs. Diagnostic sodic amphibole: *glaucophane*, imparting a distinctive blue color and schistose fabric.
- **Pseudotachylite (Dynamic Frictional Melt in Faults):** Glassy, vitreous, or crypto-crystalline material occurring as injection veins and fault-vein fault rocks, generated by extreme localized frictional heating during rapid seismic rupture ($>0.1 \text{--} 1 \text{ m/s}$) along fault planes.
- **Hornfels (Contact / Thermal Metamorphism):** Tough, fine-grained, non-schistose rocks formed in the thermal aureoles of igneous intrusions under low confining pressures. Characterized by substantial static recrystallization and polygonal/decussate textures without preferred mineral orientation.
- **Skarn (Hydrothermal Metasomatism):** Coarse-grained, zoned calc-silicate rocks resulting from intense chemical mass transfer (metasomatism) between silicate magmas/hydrothermal fluids and carbonate host rocks (limestones or dolomites), richly packed with Ca-Fe-Mg-Mn silicates (garnet, pyroxene, epidote).

texture in a sample, you may use a name such as *kyanite porphyroblast schist*. Some porphyroblasts, particularly in low-grade contact metamorphism, occur as ovoid **spots**. If such spots occur in a hornfels or a phyllite (typically as a contact metamorphic overprint over a regionally developed phyllite), the term **spotted hornfels** or **spotted phyllite** would be appropriate. Some gneisses have large eye-shaped grains (commonly feldspar) that are derived from preexisting large crystals by shear. Individual grains of this sort are called **auge** (German for *eye*), and the (German) plural is **augen**. An **augen gneiss** is a gneiss with augen structure.

Other modifying terms that we may want to add as a means of emphasizing some aspect of a rock may concern such features as grain size, color, and chemical aspects (aluminous, calcareous, mafic, felsic, etc.). As a general rule, we use these when the aspect is unusual. Obviously a *calcareous marble* or *mafic greenschist* is redundant, as is a *fine-grained slate*.

Two common prefixes that pertain to protolith are the terms ortho- and para-. **Ortho-** indicates an igneous parent, and **para-** indicates a sedimentary parent. The terms are used

5 HIGH-STRAIN ROCKS

Table 1 shows a simple scheme for classifying dynamically metamorphosed rocks associated with fault zones (including more ductile shear zones). For the moment, we shall consider the terms as purely descriptive, so that we can use Table 1 to readily identify high-strain rocks based on observed textural criteria in hand specimen and thin section. We shall discuss the processes involved in more detail in the next chapter. To name a high-strain rock, one must determine whether the rock is cohesive or whether it falls apart and then estimate the relative proportions of large clasts versus fine matrix (presumably derived mechanically from the once-larger clasts). A rock without cohesion is either a **fault breccia** or **fault gouge** (left column of Table 1). Such cohesionless rocks are typically restricted to very shallow crustal levels. Gouge is usually altered by groundwater to a clay-rich matrix.

Cohesive rocks are further distinguished by being either foliated or non-foliated, both of which are usually more fine-grained than the non-cohesive varieties. Non-foliated

TABLE 1 Classification of High-Strain Fault Zone Rocks

% fine matrix	Rocks without primary cohesion	Rocks with primary cohesion		
		Non-foliated	Foliated	Glass in matrix
50	Fault breccia	Microbreccia	Protomylonite	Pseudotachylite
70			Mylonite	
90	Fault gouge	Cataclasite	Ultramylonite	

After Higgins (1971)

Table 22-1 from John D. Winter (p. 502): Classification of high-strain fault zone rocks (after Sibson, 1977; Wise et al., 1984). Note the distinction between non-foliated cataclasites, foliated mylonites, and **pseudotachylites** (glassy, vitreous fault rocks formed by frictional melting).

cohesive rocks are either **microbreccias** (<70% clasts) or **cataclasites** (>70% clasts). Foliated cohesive rocks are **mylonites** and are subdivided as shown in the third column of Table 1. Some geologists use the term **phyllonite** to indicate a mica-rich mylonite. As with practically any other classification, all the categories in Table 1 are gradational.

Virtually all high-strain processes involve grain size reduction. The degree of recrystallization in the matrix is not a factor unless it is very advanced (perhaps aided by a later metamorphic event). The prefix **blasto-** is then added, meaning that the mylonitic texture is still apparent but largely inherited from an earlier high-strain event. In high-strain rocks, the prefix **blasto-** is usually restricted to foliated rocks because a recrystallized cataclasite may be difficult to recognize without the foliation as a clue. In extreme cases of high-strain deformation, thin, commonly anastomosing seams of glassy rock, known as **pseudotachylite** (Shand, 1916), are generated. The glass is attributed to melting due to frictional heat.

Figure 2 shows the common distribution of fault-related rock types with depth. In shallow areas, the fault zone is narrower and involves more brittle behavior, resulting in anastomosing faults containing rocks that commonly lack cohesion (breccia and gouge). Below this zone, the confining pressure forces the walls of the fault together and deformation becomes more thorough and pervasive so that the constituents become finer. The rocks thus grade downward into cataclasites and microbreccias, occasionally with pseudotachylites. Deeper yet, the rocks take on a foliated

more detail in the next chapter. The width of the shear zone increases with depth and becomes both more ductile and more evenly distributed throughout the rock matrix. At the greatest depths, the shear is distributed over a wide area and is not very intense at any single place. The rocks are ductile, and gneisses generally develop that are indistinguishable from gneisses of regional orogenic metamorphism.

The terminology for high-strain rocks is complicated by recent advances in our understanding of the deformation processes involved in their generation. The classification in Table 1 was developed before our modern understanding of the importance of recrystallization and the ductile processes that commonly accompany shear-zone deformation. It suffers from using terms that have genetic implications of purely brittle behavior. The term **cataclasis**, as originally conceived, refers to a process of mechanical crushing and granulation of a rock and its mineral constituents (with no accompanying recrystallization). This was once generally accepted as the predominant process operating in most fault and shear zones. **Cataclastic** is an adjective form used to describe the processes and textures resulting from cataclasis. Lapworth (1885) first used the term **mylonite** to describe a fine-grained laminated rock from the Moine Thrust of Scotland. The term was derived from the Greek word *mule*, for mill, meaning “to grind.” Although cataclasis is clearly the dominant process acting in shallow brittle fault zones, resulting in broken up fault gouges and breccias, this is usually not the case at even a modest depth.

Figure 2 indicates that purely mechanical grinding

Figure 22-2 from John D. Winter (p. 503): Schematic cross section through a crustal fault zone showing the depth distribution of fault rocks. Pseudotachylites originate along seismic slip surfaces in the upper to middle crust where dry, high-friction conditions permit flash frictional melting.

3.1 Shear Sense Indicators

In order to properly interpret deformation, it is useful to know the sense of shear between opposing blocks across a fault zone that is no longer active. Plate tectonics has increased our interest in knowing the sense of shear, as it may help us to determine the relative motions of crustal blocks involved in orogeny. The *direction* of motion in a shear zone is in the shear zone plane and assumed to be parallel to fault striations or mineral elongation. The *sense* of shear (which of the two opposite possible motions associated with a given direction), however, is more difficult to determine. It is important to understand shear offset in three dimensions, as the sense of offset on a two-dimensional surface may be misleading if the direction of motion has not clearly been determined. Orientation of samples is critical, and it will be assumed in the following discussion that we are observing textures in a direction that is both *in the shear plane and normal to the direction of motion*. In the figures that follow, the shear plane will thus be normal to the page and the direction of shear will be right–left (and the sense will consistently be dextral (clockwise)). Proper orientation may be difficult to manage on two-dimensional outcrops in the field but can be accomplished by cutting thin sections from appropriately reoriented hand specimens.

If a planar or linear structure is cut by the shear plane, we can use the offset or drag-curvature features to easily determine the sense of offset (as in the dark dikelet in Figure 17a, top, or the foliation in Figure 8). Many shear zones, however, do not show foliations that curve into them, at least not on the outcrop or smaller scale. When offset or curved markers are absent, careful textural analysis and interpretation are required.

Oblique foliations, such as those developed in the schematic quartz grains of Figure 17a, cut across an

Figure 23-16b from John D. Winter (p. 517): Outcrop appearance of a dark, vitreous pseudotachylite vein cutting sharply across host gneiss, showing injection veinlets extending outward into wall rock fractures.

2. Authoritative Textbook Definitions (Winter, Ch 22, pp. 500–502)

Blueschist: *"A blue-amphibole-bearing metamorphosed mafic rock. Glaucophane, the characteristic blue amphibole, indicates high pressures and low temperatures typical of subduction zones... Most blueschists are foliated (schistose) due to directed tectonic stress during subduction accretion."*

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., p. 501)

Pseudotachylite: *"A dark, very fine-grained to glassy rock that typically occurs as thin veins or intrusive veinlets in fault zones. It is formed by frictional melting of the wall rock during rapid seismic slip events on faults. When the melt chills against the cold country rock, it forms a vitreous or cryptocrystalline matrix with rounded host clasts."*

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., p. 502)

Hornfels: *"A type of granofels that is typically fine-grained, non-schistose, and occurs in contact metamorphic aureoles. Hornfelses are tough and splintery, exhibiting substantial recrystallization into equant grains without preferred orientation because deformation is absent during thermal baking."*

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., p. 500)

Skarn: *"A calc-silicate rock resulting from contact metamorphism and metasomatism of carbonate rocks by hydrothermal fluids derived from an igneous intrusion... Characterized by Ca-Fe-Mg-Mn silicates (such as grossular-andradite garnet, diopside-hedenbergite, epidote, wollastonite) that develop pronounced compositional zones and bands."*

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., p. 501)



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 7

Question Statement

Q.7: Match List-I with List-II:

List-I (Rocks)

A. Blueschist

B. Pseudotachylite

C. Hornfels

D. Skarn

List-II (Descriptions)

1. Ultrafine-grained vitreous looking material occurring as thin veins

2. Na-amphibole containing schistose rock

3. A non-schistose substantial recrystallization

4. A metasomatic Ca-Fe-Mg-Mn-silicate rock with compositional zones and bands

Correct Match: A – 2, B – 1, C – 3, D – 4



Blueschist is a Na-amphibole schist; Pseudotachylite is vitreous fault melt; Hornfels is non-schistose thermal recrystallization; Skarn is metasomatic Ca-Fe-Mg-Mn silicate rock.

Exhaustive Pairwise Critical Analysis

A → 2: Blueschist ↔ Na-amphibole containing schistose rock

Why this match is conclusive: Blueschists are defined mineralogically by the predominance of the blue sodium amphibole **glaucophane** ($\text{Na}_2(\text{Mg}_3\text{Al}_2)\text{Si}_8\text{O}_{22}(\text{OH})_2$) or crossite/riebeckite, formed by high-pressure, low-temperature metamorphism of basaltic oceanic crust in subduction zones. Because they undergo intense non-coaxial deformation within the subduction accretionary wedge, blueschists almost universally possess a pronounced tectonic foliation and are thus true *schistose rocks*.

B → 1: Pseudotachylite ↔ Ultrafine-grained vitreous looking material occurring as thin veins

Why this match is conclusive: Pseudotachylites are fossil earthquakes: they represent rock melt generated by extreme frictional dissipation along fault surfaces during seismic rupture. The generated silicate melt is violently injected into surrounding fractures and rapidly quenched against cold host rock, solidifying into an *ultrafine-grained, glassy (vitreous) or cryptocrystalline black material occurring as sharp, thin veins*.

C → 3: Hornfels ↔ A non-schistose substantial recrystallization

Why this match is conclusive: Hornfels is the classic product of thermal contact metamorphism in the static aureoles surrounding igneous plutons. Because differential stress and directed tectonic shearing are minimal or absent during thermal baking, the preexisting rock undergoes *substantial static recrystallization* to form an interlocking, equigranular polygonal or decussate mosaic that is entirely *non-schistose* (lacking foliation).

D → 4: Skarn ↔ A metasomatic Ca-Fe-Mg-Mn-silicate rock with compositional zones and bands

Why this match is conclusive: Skarns (also known as tactites) are formed by intense metasomatic mass exchange driven by hydrothermal fluids infiltrating between a silicate magma (providing Si, Fe, Al) and carbonate rocks (providing Ca, Mg, CO₂). This reactive fluid flow produces a complex suite of *Ca-Fe-Mg-Mn silicates* (grossular-andradite, diopside-hedenbergite, wollastonite, epidote, bustamite) arranged in sharp, distinct *compositional zones and reaction bands*.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Comprehensive Metamorphic Rock Summary & 8 Exam-Grade Multiple-Choice Questions

Master Comparative Synthesis of the Four Metamorphic Rock Types

Rock Type	Dominant Genetic Setting	Essential Mineralogy	Diagnostic Fabric / Texture	P-T Regime
Blueschist	Subduction Zone (Accretionary Wedge)	Glaucophane, Lawsonite, Jadeite, Aragonite	Strongly foliated, Nematoblastic / Schistose	High P (>6–10 kbar), Low T (200–450°C)
Pseudotachylite	Seismic Fault Zones (Brittle-Ductile)	Silicate glass, microlites, rounded relic clasts	Vitreous, cryptocrystalline, intrusive thin veins	Flash frictional heating (T > 1100°C locally)
Hornfels	Thermal Contact Aureoles	Cordierite, Andalusite, Biotite, Plagioclase, Quartz	Non-schistose, Granoblastic polygonal, Decussate	Low P (< 2–3 kbar), Moderate to High T
Skarn	Contact Hydrothermal Metasomatism	Andradite-Grossular, Diopside-Hedenbergite, Wollastonite	Compositionally zoned bands, coarse crystalloblastic	Variable P-T, active chemical mass transfer

8 High-Yield Practice Questions on Metamorphic Rock Nomenclature

1. Glaucophane, the defining sodium amphibole in blueschists, belongs to which mineral group formula?

- (a) $\text{Ca}_2(\text{Mg,Fe})_5\text{Si}_8\text{O}_{22}(\text{OH})_2$
- (b) $\text{Na}_2\text{Mg}_3\text{Al}_2\text{Si}_8\text{O}_{22}(\text{OH})_2$
- (c) $(\text{Mg,Fe})_7\text{Si}_8\text{O}_{22}(\text{OH})_2$
- (d) $\text{KAl}_2\text{Si}_3\text{AlO}_{10}(\text{OH})_2$

Answer: (b) — Glaucophane is a sodic amphibole with the end-member formula $\text{Na}_2\text{Mg}_3\text{Al}_2\text{Si}_8\text{O}_{22}(\text{OH})_2$.

2. Pseudotachylites are distinguished from mylonites primarily because pseudotachylites form by:

- (a) Ductile crystal-plastic dislocation creep
- (b) Frictional melting during rapid seismic slip
- (c) Slow pressure solution mass transfer
- (d) Static thermal annealing in an aureole

Answer: (b) Frictional melting during rapid seismic slip — Mylonites form by ductile plastic flow without melting, whereas pseudotachylites are generated by flash frictional melting during fast seismic slip events.

3. Contact metamorphic aureoles surrounding shallow granitic intrusions typically produce which non-foliated rock from pelitic shale protoliths?

- (a) Mylonite
- (b) Pelitic Hornfels
- (c) Blueschist
- (d) Gneiss

Answer: (b) Pelitic Hornfels — Shales baked under static conditions in thermal aureoles transform into dense, fine-grained, non-schistose hornfelses (spotted hornfels with cordierite/andalusite).

4. An "exoskarn" refers to a metasomatic skarn developed within:

- (a) The intrusive igneous body itself
- (b) The sedimentary carbonate host rock adjoining the pluton
- (c) Ocean floor basaltic pillow lavas
- (d) Mantle peridotite xenoliths

Answer: (b) — Exoskarn forms in the surrounding sedimentary carbonate country rock, whereas endoskarn forms by replacement within the igneous intrusion itself.

5. Under polarizing petrographic microscope, the glassy matrix of an unaltered pseudotachylite appears:

- (a) Strongly pleochroic
- (b) Completely isotropic (dark under crossed polars)
- (c) Highly birefringent
- (d) Lamellar twinned

Answer: (b) Completely isotropic (dark under crossed polars) — True volcanic and frictional glass lacks crystalline order and is optically isotropic.

6. Which diagnostic mineral pair in blueschists indicates the complete breakdown of plagioclase at high pressure?

- (a) Albite + Anorthite
- (b) Jadeite + Quartz
- (c) Calcite + Quartz
- (d) Chlorite + Biotite

Answer: (b) Jadeite + Quartz — Albite breaks down at high pressure via the univariant reaction: Albite = Jadeite + Quartz, a definitive high-pressure index equilibrium.

7. The decussate texture often observed in hornfelses is characterized by:

- (a) Parallel aligned mica flakes defining continuous schistosity
- (b) Randomly oriented, interlocking elongate or platy crystals lacking preferred orientation
- (c) Alternating felsic and mafic compositional banding
- (d) Rounded porphyroclasts with recrystallized tails

Answer: (b) — Decussate texture consists of interlocking prismatic or platy grains (e.g., micas, amphiboles) oriented completely at random due to absence of deviatoric stress.

8. Skarn deposits are world-class economic sources for which metals?

- (a) Fe, Cu, W, Sn, Au, Zn, Pb
- (b) Exclusively alkali metals like Li and Na
- (c) Noble gases
- (d) Halite and sylvite

Answer: (a) Fe, Cu, W, Sn, Au, Zn, Pb — Metasomatic skarns host major hydrothermal ore deposits of iron (magnetite skarn), copper, tungsten (scheelite skarn), gold, and lead-zinc.